**6.** (i) The circle $C_1$ has equation

$$x^2 + y^2 + 10x - 12y = k \qquad \text{where } k \text{ is a constant}$$

(a) Find the coordinates of the centre of $C_1$

**(2)**

(b) State the possible range in values for $k$.

**(2)**

(ii) The point $P(p, 0)$, the point $Q(-2, 10)$ and the point $R(8, -14)$ lie on a different circle, $C_2$

Given that

- $p$ is a positive constant
- $QR$ is a diameter of $C_2$

find the exact value of $p$.

**(4)**